Power Meter

Technical Datasheet

Voltage, Current, Power and Energy Measurement System

Version 1.13

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Contents

1 General Description

The Power Meter is a standalone system for measuring and recording electrical quantities, designed primarily for monitoring photovoltaic sources and other low-power electrical systems. It measures voltage and current through the INA219 module and calculates instantaneous power as well as the accumulated energy since the most recent system startup or reset.

Measurements are displayed locally on an OLED screen and stored on a microSD card in CSV format for subsequent processing and analysis. At the same time, three LED indicators provide immediate visual information about the overall system status, successful measurement acquisition, and the occurrence of an error.

The system is used for the experimental evaluation of photovoltaic configurations, allowing objective comparison of different operating conditions or different electrical configurations. Continuous logging enables long-duration measurements without requiring a permanent connection to a computer.

2 Technical Specifications

The main technical specifications of the Power Meter are presented in Table 1.

Characteristic	Specification
Main unit	Arduino Pro Mini 5 V / 16 MHz
Microcontroller	ATmega328P
Measurement sensor	INA219 breakout with on-board shunt resistor
Measured / calculated quantities	Photovoltaic source voltage, current, instantaneous power and accumulated energy
Units	V, mA, mW and mWh
Nominal measurement interval	100 ms
Nominal measurement rate	approximately 10 Hz
OLED	SSD1306, 0.96", 128×64 pixels, I^2C
OLED address	`0x3C`
INA219 address of the specific module	`0x45`
I^2C	SDA=A4, SCL=A5
microSD	SPI: CS=D10, MOSI=D11, MISO=D12, SCK=D13
Log file	`DATA.csv`
CSV fields	`Time_ms`, `Voltage_V`, `Current_mA`, `Power_mW`, `Energy_mWh`
Serial communication	9600 baud
Power Meter supply	Regulated 5 V DC at the VCC pin
Visual indicators	D2: System OK, D3: Measurement OK, D4: Measurement Error
Programming	USB-to-TTL serial converter, e.g. FTDI breakout board
Reference test panel	9 V / 2 W
Reference load for one panel	51 Ω / 5 W, for the specific 9 V / 2 W test

Table 1: Main technical specifications of the Power Meter.

Important note regarding the I^2C address: The value `0x45` applies to the specific INA219 module used in this implementation and has been confirmed with an I^2C scanner. If the module is replaced, the address must be checked again and the firmware must use the same value.

Metrological limits: The maximum allowable voltage, current and power limits are not stated here as general values because they depend on the specific INA219 breakout board, the shunt resistor, and the library configuration. Before use in a different configuration, the specifications of the specific module must be checked and must not be exceeded.

3 System Hardware

The Power Meter is implemented using measurement, processing, display and data storage modules. The central operation is based on the Arduino Pro Mini, which collects measurements from the INA219, calculates power and energy, and manages the OLED display, the microSD card and the LED indicators.

3.1 Arduino Pro Mini

The Arduino Pro Mini is the main control unit. This implementation uses the 5 V / 16 MHz version with the ATmega328P microcontroller.

Programming from a computer requires an external USB-to-TTL serial converter, such as an FTDI breakout board, because the Arduino Pro Mini does not include an integrated USB-to-serial converter.

Pins A4 and A5 are used as SDA and SCL, respectively. On several Arduino Pro Mini versions they are located on internal pads or outside the main row of pin pads, so their physical location must be confirmed before soldering.

3.2 INA219

The INA219 module is used to measure the voltage and current of the circuit under test. The specific module used in this implementation communicates with the Arduino through I^2C at address `0x45`.

The INA219 breakout board used here includes a shunt resistor directly on the board. Therefore, this implementation does not require a separate external shunt resistor for current measurement.

The 51 Ω load resistor is a different component and must not be confused with the shunt resistor.

3.3 OLED Display

For local display, an SSD1306 0.96” OLED with a resolution of 128×64 pixels and I^2C communication at address `0x3C` is used.

The OLED and the INA219 share the same I^2C bus and are connected to the common SDA and SCL lines. Simultaneous operation is possible because they use different addresses.

3.4 microSD Module

The microSD module is used for permanent storage of the measurements and communicates with the Arduino through SPI. The log file is named `DATA.csv`.

The following are required:

- a microSD module compatible with the supply voltage and logic levels of the Arduino Pro Mini 5 V,
- a properly formatted microSD card,
- an external SD/microSD card reader for transferring the data to a computer.

The microSD module must be powered according to the specifications of the specific module. If the card itself or the module operates internally at 3.3 V, appropriate power regulation and, where required, logic-level adaptation must be provided. It must not be assumed that every microSD module is safe for direct 5 V logic signals.

3.5 LED Indicators

The Power Meter includes three LED indicators:

- **D2 – System OK:** remains ON only when the INA219 and microSD are connected and operating correctly. If either one is unavailable at startup or becomes disconnected during operation, the LED turns OFF and remains OFF until reset or removal of power,
- **D3 – Measurement OK:** turns ON momentarily for approximately 40 ms after a valid measurement,
- **D4 – Measurement Error:** turns ON after a failed measurement, when the INA219 or microSD is unavailable at startup, or when either one becomes disconnected during operation. It remains ON until reset or removal of power.

Each LED is connected through a 330 Ω resistor for current limiting.

3.6 Power Supply

The Power Meter requires a regulated 5 V DC supply. Because the 5 V version of the Arduino Pro Mini is used, the regulated voltage is applied to the **VCC** pin and not to RAW.

The Power Meter supply is independent of the electrical energy produced by the photovoltaic panel. The supply ground, the Power Meter ground and the negative terminal of the photovoltaic panel are connected to a common electrical reference point.

3.7 Assembly Materials

The final assembly additionally requires:

- perforated prototyping board or a suitable PCB,
- connection wires,
- pin headers and suitable connectors,
- 330 Ω resistors for the LEDs,
- power-supply cables,
- crocodile clips or suitable terminals for external connections,
- soldering materials.

3.8 External Measurement Load

An electrical load is required to measure a photovoltaic panel so that the panel operates in a closed circuit and supplies current. The load is not part of the INA219 sensor and is not the shunt resistor.

In the reference experimental setup with **one** 9 V / 2 W photovoltaic panel, a 51 Ω load resistor is used.

At 9 V, the theoretical power in the resistor is:

$$P_R = \frac{V^2}{R} = \frac{(9\,\mathrm{V})^2}{51\,\Omega} \approx 1.59\,\mathrm{W}$$

For the experimental implementation, a **51 Ω / 5 W** resistor is recommended so that adequate thermal margin is available.

Caution: The 51 Ω / 5 W value applies only to the reference configuration described above. For two panels in series, two panels in parallel, or any other photovoltaic configuration, the appropriate load and required power rating must be recalculated from the electrical characteristics of the specific configuration.

4 System Electronic Schematics

This section presents the electrical architecture of the Power Meter and the internal schematic of the Arduino Pro Mini.

Particular attention is required to distinguish the INA219 supply connection from the photovoltaic power path. The INA219 VCC pin powers only the sensor electronics, while the IN+ and IN− terminals form the current measurement path.

$$\boxed{\begin{aligned} 5\,\mathrm{V} &\longrightarrow \text{INA219 VCC} \\ \text{PV+} &\longrightarrow \text{INA219 IN+} \\ \text{INA219 IN−} &\longrightarrow \text{LOAD} \longrightarrow \text{PV−} \end{aligned}}$$

4.1 Power Meter Interconnection Diagram

The main unit of the system is the Arduino Pro Mini. The INA219 and OLED use the $\mathrm{I^2C}$ bus on A4/A5, the microSD uses the hardware SPI bus on D10–D13, and the three LEDs use digital outputs D2–D4.

Figure 1 presents the complete electronic schematic of the implementation. In the photovoltaic power path, the positive PV+ terminal is connected to **IN+** of the INA219 module. From **IN−**, the current flows to the load resistor and returns to the negative PV− terminal.

Important clarification: The label at the INA219 IN+ input must be **PV+**, not VCC. VCC is a separate pin and is connected to the 5 V Power Meter supply.

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Required path:

5 V → INA219 VCC

PV+ → INA219 IN+

INA219 IN– → LOAD → PV–

Figure 1: Complete electronic schematic of the Power Meter. The label at the INA219 IN+ input is PV+, while the INA219 VCC is connected independently to the 5 V supply.

4.2 Arduino Pro Mini Internal Schematic

The Arduino Pro Mini is based on the ATmega328P. For the Power Meter, pins A4 and A5 are particularly important because they correspond to the SDA and SCL functions of the hardware I^2C bus.

Similarly, D11, D12 and D13 are the MOSI, MISO and SCK lines of the hardware SPI bus, while D10 is used as the Chip Select for the microSD. D2, D3 and D4 are used for the three LED indicators.

Note regarding A4/A5: On the Arduino Pro Mini, A4 and A5 may be located on separate internal pads rather than on the main pin row. Their position must be confirmed from the schematic/pinout of the specific board.

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or
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Figure 2: Internal electronic schematic of the Arduino Pro Mini and mapping of its pins to the ATmega328P.

5 Wiring

Correct wiring is essential both for reliable measurement and for safe operation of the system.

The Power Meter is powered by an independent regulated 5 V source, while the photovoltaic panel is the electrical-energy source being measured.

5.1 Photovoltaic Panel and Load Connection

The current path for the reference setup is:

$$\boxed{\text{PV+} \rightarrow \text{INA219 IN+} \rightarrow \text{INA219 IN–} \rightarrow 51\,\Omega \rightarrow \text{PV–}}$$

The positive terminal of the photovoltaic panel is connected to IN+ of the INA219.

IN– of the INA219 is connected to one side of the 51 Ω load resistor, and the other side of the resistor returns to the negative terminal of the photovoltaic panel.

The INA219 VCC is not part of this power path. VCC powers only the electronics of the INA219 module from the Power Meter 5 V supply.

5.2 Common Ground

The negative PV– terminal is connected to the common GND of the Power Meter.

The following are connected to the same electrical reference point:

- Arduino Pro Mini GND,
- INA219 GND,
- OLED GND,
- microSD GND,
- GND of the independent 5 V supply,
- the return path of the load to PV–.

5.3 Power Meter Supply Connection

The regulated 5 V supply is applied to the VCC pin of the 5 V Arduino Pro Mini. The negative supply terminal is connected to the common GND.

The RAW input is not used when regulated 5 V is already being provided.

5.4 INA219 and OLED Display Connection

The INA219 and OLED share the SDA and SCL lines:

Device	Signal	Arduino Pro Mini
INA219	SDA	A4
INA219	SCL	A5
OLED SSD1306	SDA	A4
OLED SSD1306	SCL	A5

Table 2: I^2C connections.

The addresses used in this implementation are `0x45` for the INA219 and `0x3C` for the OLED.

The same values must also be used in the firmware.

5.5 microSD Module Connection

microSD	Arduino Pro Mini
CS	D10
MOSI	D11
MISO	D12
SCK	D13

Table 3: SPI connections of the microSD module.

D11–D13 are the hardware SPI lines of the ATmega328P. D10 is used as Chip Select.

5.6 LED Indicator Connection

LED	Pin	State	Function
System OK	D2	ON/OFF	ON only when INA219 and SD are available. If either one fails or is disconnected, it remains OFF until reset or removal of power
Measurement OK	D3	pulse	approximately 40 ms after a valid sample
Measurement Error	D4	latched ON	remains ON after a failed measurement or loss of INA219/SD until reset or removal of power

Table 4: Operation of the LED indicators.

Each LED is connected through a $330\,\Omega$ resistor.

5.7 USB-to-TTL / FTDI Connection for Programming

A USB-to-TTL converter is used for programming and for the Serial Monitor.

The basic connections are:

USB-to-TTL / FTDI	Arduino Pro Mini
TX	RX
RX	TX
GND	GND
DTR	DTR / reset circuit, if supported
VCC	VCC only when the converter is used to provide power

Table 5: Basic USB-to-TTL / FTDI connection.

When the Power Meter is already powered from an independent regulated 5 V source, the VCC line of the USB-to-TTL converter must not also be connected unless it has been confirmed that the two sources can be safely connected together.

GND must remain common.

5.8 External Wiring Diagram

The complete external wiring of the system is presented in Figure 3.

The diagram shows the connection of the photovoltaic panel, the external load, the Power Meter and the independent regulated 5 V supply.

The positive terminal of the photovoltaic panel is connected to the INA219 IN+ input. The INA219 IN– output is connected to the external load, and the other side of the load returns to the negative terminal of the photovoltaic panel.

The negative terminal of the photovoltaic panel is also connected to the common ground point of the Power Meter.

In this way, the photovoltaic panel, the measurement circuit and the external power supply use a common electrical reference point.

The Power Meter is supplied independently by a regulated 5 V DC source. The positive supply terminal is connected to the VCC of the Arduino Pro Mini, while the negative supply terminal is connected to the common GND.

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Figure 3: External wiring of the Power Meter with the photovoltaic panel, the external load and the independent 5 V supply. The load return path and the common ground point of the system are also shown.

The current path in the measurement setup is:

$$\boxed{\text{PV+} \rightarrow \text{INA219 IN+} \rightarrow \text{INA219 IN–} \rightarrow \text{LOAD} \rightarrow \text{PV–}}$$

For the reference setup with one 9 V / 2 W photovoltaic panel, the load is a 51 Ω / 5 W resistor.

6 Operation

After power is applied, the Arduino Pro Mini initializes the I^2C bus, the INA219 module, the OLED and the microSD.

It then performs repeated measurement cycles with a nominal interval of 100 ms.

6.1 Startup and Initialization

During startup:

1. D2, D3 and D4 are initialized and initially set to LOW,
2. serial communication starts at 9600 baud,
3. the hardware I^2C bus starts on A4/A5,
4. the INA219 is checked and initialized at address `0x45`,
5. the OLED is checked and initialized at address `0x3C`,
6. the microSD is checked through D10 and `DATA.csv` is created only if it does not already exist,
7. the System OK LED is updated.

If a device is unavailable, the system continues the functions that remain possible.

The System OK LED remains ON only when both the INA219 and the microSD are available. If either one is unavailable at startup, the LED remains OFF until reset or removal of power.

6.2 Measurement Procedure

The nominal time between consecutive cycles is 100 ms:

$$f_{\text{nom}} = \frac{1}{0.1\,\text{s}} = 10\,\text{Hz}$$

The actual time between two cycles is measured through `millis()` and used as `elapsedMillis`.

Therefore, it is not assumed to be exactly 100 ms, because updating the OLED, writing to the SD card and generating the LED pulse require additional execution time.

Before each sample, the code checks the INA219. If the sensor had been disconnected and then reappears, a reconnection attempt is made.

6.3 Voltage and Power Calculation

The INA219 returns the bus voltage, the voltage drop across the shunt, and the current.

With the wiring used in this implementation, the photovoltaic source voltage on the IN+ side is calculated as:

$$V_{\mathrm{PV}} = V_{\mathrm{bus}} + \frac{V_{\mathrm{shunt,mV}}}{1000}$$

This value is stored in the `Voltage_V` field.

The instantaneous electrical power of the measured photovoltaic source is calculated from:

$$P_{\mathrm{mW}} = V_{\mathrm{PV,V}} \cdot I_{\mathrm{mA}}$$

because:

$$\mathrm{V} \times \mathrm{mA} = \mathrm{mW}$$

Clarification: The calculated voltage above is the voltage on the IN+ side and not only the voltage across the load resistor. For this reason, this datasheet uses the notation V_{PV} instead of the ambiguous V_{load}.

6.4 Accumulated Energy Calculation

The `Energy_mWh` variable represents the accumulated energy since the most recent Arduino startup or reset.

By itself, it is not a persistent energy meter across separate restarts.

For each valid measurement:

$$\Delta E_{\mathrm{mWh}} = P_{\mathrm{mW}} \cdot \frac{\Delta t_{\mathrm{ms}}}{3\,600\,000}$$

and:

$$E_{\mathrm{new}} = E_{\mathrm{old}} + \Delta E$$

The integration uses the actual time Δt between samples.

6.5 Measurement Display

The OLED displays the basic Voltage, Current, Power and Energy values.

The display operates without requiring a connection to the Serial Monitor.

If the OLED is disconnected during operation, the remaining functions can continue, but the current firmware version does not perform a full automatic reinitialization of the OLED without a reset.

6.6 Logging to the microSD Card

Logging is performed in `DATA.csv`.

If the file already exists at startup, it is not deleted. New rows are appended to the end of the existing file.

Each row has the form:

```
Time_ms,Voltage_V,Current_mA,Power_mW,Energy_mWh
```

For a successful measurement, all numerical values are stored.

For a failed measurement, the instantaneous voltage, current and power values may be logged as `NA`, while the energy already accumulated is retained.

If the SD card becomes unavailable during operation, the current firmware version stops writing to the SD card. The other available functions can continue, but a reset is required for full reinitialization of the SD card.

6.7 LED Indicators

LED D2 indicates the status of the INA219 and the microSD. It remains ON only while both are connected and operating correctly. If the INA219 or microSD is unavailable at startup or becomes disconnected during operation, D2 turns OFF and remains OFF until reset or removal of power.

D3 generates a short pulse after every valid measurement.

D4 operates as a latched error indicator: it turns ON after a failed measurement, when the INA219 or microSD is unavailable at startup, or when either one becomes disconnected during operation. The status of the INA219 and microSD is checked after each measurement. D4 remains ON until reset or removal of power, even if the cause of the error is later restored.

6.8 Continuous Operation

The operating cycle repeats continuously:

INA219 check → measurement → energy calculation → LED → Serial → OLED → SD

This process enables long-duration measurements and the creation of a production history for subsequent analysis.

7 Functional Verification and Acceptance Criteria

Before being integrated into the final experimental photovoltaic setup, the complete Power Meter board was subjected to a series of eleven tests covering operation, reliability and data integrity.

The purpose of the tests is not to provide formal metrological certification, but to technically verify that this specific implementation operates in a repeatable and predictable manner, that its measurements are consistent with a reference instrument, that data are stored correctly, and that basic fault conditions are handled without loss of system control.

The final board successfully completed the tests below and, based on the respective acceptance criteria, is considered **suitable for use in the present photovoltaic project**.

7.1 Power-on Test

Repeated Power Meter power-on and power-off cycles were performed.

Acceptance criterion:

- normal startup of the Arduino Pro Mini,
- correct detection of the INA219, OLED and microSD,
- no program freeze during startup,
- correct start of the measurement and logging process.

Result: PASS

7.2 Voltage Accuracy Test

The voltage recorded by the Power Meter was compared with the reading of a reference multimeter at different operating points.

The relative deviation is calculated from:

$$\varepsilon_V(\%) = \frac{|V_{\mathrm{PM}} - V_{\mathrm{ref}}|}{|V_{\mathrm{ref}}|} \cdot 100$$

Acceptance criterion:

$$\boxed{\varepsilon_V \leq 2\%}$$

Result: PASS

7.3 Current Accuracy Test

The current recorded by the Power Meter was compared with a reference multimeter under different load conditions.

The relative deviation is calculated from:

$$\varepsilon_I(\%) = \frac{|I_{\mathrm{PM}} - I_{\mathrm{ref}}|}{|I_{\mathrm{ref}}|} \cdot 100$$

Acceptance criterion:

$$\boxed{\varepsilon_I \leq 3\%}$$

Result: PASS

7.4 Instantaneous Power Test

The reference power was calculated from the voltage and current measurements of the multimeter:

$$P_{\text{ref}} = V_{\text{ref}} \cdot I_{\text{ref}}$$

and compared with the instantaneous power recorded by the Power Meter.

The relative deviation is calculated from:

$$\varepsilon_P(\%) = \frac{|P_{\text{PM}} - P_{\text{ref}}|}{|P_{\text{ref}}|} \cdot 100$$

Acceptance criterion:

$$\boxed{\varepsilon_P \leq 4\%}$$

Result: PASS

7.5 Energy Calculation Test

The system operated with as constant a load as possible for approximately one hour. The expected energy was calculated from:

$$E_{\text{ref}} = P_{\text{ref}} \cdot t$$

and compared with the energy recorded by the Power Meter.

The relative deviation is calculated from:

$$\varepsilon_E(\%) = \frac{|E_{\text{PM}} - E_{\text{ref}}|}{|E_{\text{ref}}|} \cdot 100$$

Acceptance criterion:

$$\boxed{\varepsilon_E \leq 5\%}$$

Result: PASS

7.6 `DATA.csv` Integrity Test

The `DATA.csv` file was checked after logging to confirm that it can be read and used normally for subsequent processing.

Acceptance criterion:

- the file opens without error,

- there are no corrupted or unreadable rows,
- the numerical values remain reasonable and consistent,
- the relationship
$$P_{\mathrm{mW}} \approx V_{\mathrm{V}} \cdot I_{\mathrm{mA}},$$
is verified on a representative basis,
- the accumulated energy shows a reasonable progression within each measurement session.

Result: PASS

7.7 INA219 Fault Test

The behavior of the system was tested when the INA219 was unavailable or was disconnected during operation.

Acceptance criterion:

- the Arduino does not freeze,
- the absence of the INA219 is detected by the firmware,
- the System OK LED turns OFF,
- the Measurement Error LED turns ON and remains ON until reset or removal of power,
- invalid measurements are not recorded as normal data when the sensor is unavailable.

Result: PASS

7.8 SD Card Fault Test

The behavior of the system was tested without an available microSD card or after its disconnection during operation.

Acceptance criterion:

- the Arduino does not freeze and does not enter an uncontrolled state,
- the absence of the microSD is correctly detected,
- the System OK LED turns OFF,
- the Measurement Error LED turns ON and remains ON until reset or removal of power,
- functions that do not depend on SD storage continue to operate wherever possible.

Result: PASS

7.9 Real Test with a Photovoltaic Panel

The Power Meter was tested with a real photovoltaic panel and the specified electrical load of the experimental setup.

System operation was evaluated under real changes in illumination and generated electrical power.

Acceptance criterion:

- stable system operation with a real photovoltaic source,
- expected variation of voltage, current and power when illumination changes,
- agreement of the measurements with the corresponding reference measurements,
- normal display of the measurements on the OLED,
- normal storage of the data on the microSD.

Result: PASS

7.10 24-hour Endurance & Sampling Test

Continuous operation and logging were performed for a period longer than 24 hours in order to evaluate the behavior of the final board under conditions corresponding to actual experimental use.

Acceptance criterion:

- `DATA.csv` remains readable and free of corruption,
- the INA219 and microSD remain operational throughout the test,
- measurement logging is continuous,
- the sampling interval remains close to the intended system operation without large unexplained data gaps,
- no excessive heating or other indication of electrical instability appears,
- a possible Arduino restart can be identified from the restart of the `Time_ms` value and handled correctly during subsequent processing.

`DATA.csv` is not deleted after an Arduino restart. New measurements are appended to the existing file, creating a new measurement session.

During subsequent processing, the Python program identifies the different sessions through the restart of the `Time_ms` value and includes the time and energy of each session in the calculations.

In this way, an Arduino restart does not cause loss of measurements already stored on the microSD card and does not prevent calculation of the total duration and total energy of the test.

Result: PASS

7.11 Final Power-cycle Test

After completion of the long-duration test, additional power-on and power-off cycles of the complete assembly were performed.

This test confirms that long-duration continuous operation does not affect the board's ability to restart and return to normal operation.

Acceptance criterion:

- correct Arduino restart,
- detection of the INA219, OLED and microSD,
- normal acquisition of new measurements,
- normal display of the measurements on the OLED,
- creation of new valid entries in `DATA.csv`.

Result: PASS

7.12 Overall Test Results

The acceptance-test results for the final board are summarized in Table 6.

No.	Test	Result
1	Power-on Test	**PASS**
2	Voltage accuracy test	**PASS**
3	Current accuracy test	**PASS**
4	Instantaneous power test	**PASS**
5	Energy calculation test	**PASS**
6	`DATA.csv` integrity test	**PASS**
7	INA219 Fault Test	**PASS**
8	SD Card Fault Test	**PASS**
9	Real test with a photovoltaic panel	**PASS**
10	24-hour Endurance & Sampling Test	**PASS**
11	Final Power-cycle Test	**PASS**

Table 6: Summary of the acceptance-test results for the final board.

7.13 Final Board Evaluation

Successful completion of all eleven tests provides technical documentation that this specific Power Meter board is functional, stable and suitable for collecting the experimental data of the present project.

The reliability of the implementation is not based only on correct startup or the display of measurements. Operation of the complete measurement and logging chain has been tested:

PV → INA219 → Arduino → calculation → OLED → microSD → CSV → Python

The tests include verification of the measured electrical quantities, verification of the power and energy calculations, data-integrity checks, fault tests for the main peripherals, operation with a real photovoltaic panel, and continuous operation for more than 24 hours.

In addition, it has been confirmed that data already stored on the microSD card are retained after a possible Arduino restart and can be combined during subsequent processing through the Python program.

Based on the results above, the final board is considered **successfully verified for the requirements of this specific experimental system** and can be integrated into the photovoltaic project as the main measurement and logging system.

The evaluation above applies to this specific implementation, wiring arrangement and operating conditions under which the tests were performed, and does not constitute formal certification of a metrological instrument.

8 Data Logging and Processing

8.1 CSV File Structure

The `DATA.csv` file contains five columns:

Field	**Unit**	**Source**	**Description**
`Time_ms`	ms	`millis()`	relative time since the most recent startup / reset
`Voltage_V`	V	INA219 + calculation	photovoltaic source voltage on the IN+ side
`Current_mA`	mA	INA219	measured current
`Power_mW`	mW	calculation	instantaneous power $V_{\mathrm{PV}} \cdot I$
`Energy_mWh`	mWh	integration	accumulated energy of the current session

Table 7: Fields of the `DATA.csv` file.

8.2 Multiple Startups in the Same File

The firmware does not delete `DATA.csv` when the file already exists. Therefore, after a reset or a new power-up, new rows continue to be appended to the same file.

At the same time, `millis()` and `Energy_mWh` start again from zero.

For this reason, the same CSV file may contain multiple different measurement sessions.

For tests significantly shorter than approximately 49.7 days, a new session can be identified when the new `Time_ms` value is lower than the previous one.

For much longer continuous operation, it must be taken into account that the 32-bit `millis()` value wraps around, which also causes a decrease in `Time_ms` without a reset having occurred.

The total energy of multiple sessions is calculated during subsequent data processing.

8.3 Data Processing with Python

After logging is completed, the `DATA.csv` file is transferred to a computer and processed using a dedicated program written in Python. The program is used to calculate the total duration of the measurement, the total generated energy and the average power over the entire recording period.

The processing is based mainly on the `Time_ms` and `Energy_mWh` columns of the CSV file. Voltage, current and instantaneous power values remain available in the file for further analysis, but they are not required for the calculation of total energy by this specific program.

Of particular importance is the ability to identify a possible Arduino restart. After a reset or loss and restoration of power, the `millis()` function starts again from zero, while the `Energy_mWh` variable also begins a new energy accumulation.

The program examines the values in the `Time_ms` column sequentially. If, for two consecutive entries,

$$t_{\text{new}} < t_{\text{previous}},$$

then a new measurement session is considered to have started because of an Arduino restart.

Before moving to the next session, the program adds the last recorded energy value and the final time value of the previous session to the corresponding total quantities.

If the file contains N different measurement sessions, total energy is calculated as:

$$E_{\text{total,mWh}} = \sum_{i=1}^{N} E_{i,\text{final}}$$

where $E_{i,\text{final}}$ is the last recorded `Energy_mWh` value of each session.

Similarly, the total recording duration is calculated from:

$$t_{\text{total,ms}} = \sum_{i=1}^{N} t_{i,\text{final}}$$

where $t_{i,\text{final}}$ is the final `Time_ms` value of the corresponding session.

In this way, an Arduino restart does not result in loss of the energy already recorded in `DATA.csv`, because each session is taken into account separately during subsequent processing.

After calculating the total energy, conversion from mWh to Wh is performed:

$$E_{\text{total,Wh}} = \frac{E_{\text{total,mWh}}}{1000}$$

The total time is also converted from milliseconds to seconds, minutes and hours:

$$t_{\text{s}} = \frac{t_{\text{ms}}}{1000}$$

$$t_{\text{min}} = \frac{t_{\text{s}}}{60}$$

$$t_{\text{h}} = \frac{t_{\text{s}}}{3600}$$

The generated energy per hour is then calculated:

$$E_{\text{per hour}} = \frac{E_{\text{total,Wh}}}{t_{\text{total,h}}}$$

with unit Wh/h.

Because:

$$1\,\text{Wh/h} = 1\,\text{W},$$

the quantity above is numerically equal to the average electrical power over the entire measurement period:

$$P_{\text{average}} = \frac{E_{\text{total,Wh}}}{t_{\text{total,h}}}$$

Similarly:

$$1\,\text{mWh/h} = 1\,\text{mW}.$$

For this reason, the program displays the final results both in Wh and W and in mWh and mW.

Specifically, after processing, the following are displayed:

- total energy in Wh and mWh,
- total recording duration in seconds, minutes and hours,
- generated energy per hour in Wh/h and mWh/h,
- average power in W and mW.

This method makes it possible to process a single `DATA.csv` file even when it contains more than one Power Meter operating session. Therefore, an Arduino restart does not require a separate file and does not prevent calculation of the total energy and total duration of the test.

8.4 Subsequent Analysis

After a test is completed, the microSD card is removed while the Power Meter is powered OFF and `DATA.csv` is transferred to a computer.

The basic processing of the file is performed through the Python program described above. The data can also be used in a spreadsheet, plotting software or another suitable tool for further analysis of the measured quantities.

For comparison of different photovoltaic configurations, the same units, the same calculation method and, as far as possible, comparable environmental conditions should be maintained.

9 Limitations and Safe-Use Guidelines

- The regulated 5 V supply is connected to VCC of the 5 V Arduino Pro Mini and not to RAW.
- The electrical limits of the specific INA219 module or shunt resistor must not be exceeded.
- The 51 Ω / 5 W load value applies only to the reference setup with one 9 V / 2 W panel and does not automatically apply to other configurations.
- The load resistor must be positioned so that it can dissipate the heat it generates.
- The polarity of the photovoltaic panel and the external power supply must be checked before every connection.
- The microSD card must not be removed while data are being written. The safest practice is to remove it only after power to the Power Meter has been disconnected.
- The microSD and OLED must be powered only according to the specifications of the specific modules.
- The Power Meter does not include a Real-Time Clock (RTC). `Time_ms` is relative time since the most recent startup and is not a calendar date or time of day.
- `Energy_mWh` is stored in RAM and resets to zero after a reset or loss of power. `DATA.csv` remains on the card.
- The INA219 can be checked again and reconnected by the firmware, whereas the current version does not fully and automatically reinitialize the OLED and SD after a failure.
- Address `0x45` applies to the specific INA219 module. If the sensor is replaced, a new I^2C check and corresponding firmware configuration are required.
- The Power Meter is a prototype experimental implementation. Measurements should be verified with a reference instrument before being used as calibration data or certified measurements.

10 Sources and Technical Documentation

The following reference sources are used for the design, wiring and technical documentation of the Power Meter:

1. **SparkFun Electronics**, *Arduino Pro Mini 328 – 5V/16MHz: Schematic and Hardware Documentation.* This source is used for the internal electronic schematic, pinout and pin mapping of the Arduino Pro Mini.

2. **Microchip Technology**, *ATmega328P Microcontroller Datasheet.* Used as a technical reference for pin functions, the I^2C/TWI bus, SPI and microcontroller timing.

3. **Texas Instruments**, *INA219 – Zerø-Drift, Bidirectional Current/Power Monitor With I^2C Interface, Datasheet.* Used for sensor operation and for bus-voltage, shunt-voltage and current measurement.

4. **Solomon Systech**, *SSD1306 OLED/PLED Segment/Common Driver with Controller, Datasheet.* Used as a technical reference for the OLED display controller.

5. **Arduino Documentation**, technical documentation for `Wire`, SPI, serial communication and `millis()`.

6. **SdFat Library Documentation**, documentation for accessing, creating and writing files on SD cards through Arduino microcontrollers.

Note regarding the figures: Figure 1 refers to the specific implementation of the Power Meter. Figure 2 is a reference electronic schematic of the Arduino Pro Mini board and retains the source and licensing information shown in the original drawing.